

Muhammad Ibrahim Khan

Centre for Computational Science and Mathematical Modelling, Coventry University, UK
+44 7470 800092 | khanm442@uni.coventry.ac.uk | [website](#)
[Google Scholar](#) | [GitHub](#) | [LinkedIn](#)

EDUCATION

- **Coventry University** Sep 2024 – Mar 2028 (expected)
Doctor of Philosophy (PhD), Reinforcement Learning Coventry, United Kingdom
 - Thesis: *Predictive Control Through Monte Carlo Tree Search*. Supervisor: Prof. James Brusey.
 - Developing reinforcement learning methods for energy-efficient control of building heating systems, evaluated on the BOPTEST benchmark suite under dynamic electricity pricing and deadline constraints.
 - Related strands: interpretability of learned control policies, and the numerical stability of policy-gradient methods.
 - Fully funded studentship, Centre for Computational Science and Mathematical Modelling (CSMM).
- **National University of Sciences and Technology (NUST)** Sep 2019 – Jun 2023
BEng Mechanical Engineering Islamabad, Pakistan
 - Capstone: *Autonomous Weeding Robot* — vision-based weed detection, ROS navigation, and laser-based weed removal.

PUBLICATIONS

Peer-reviewed

- **Khan, M.I., et al. Deadline-Aware, Energy-Efficient Control of Domestic Immersion Hot Water Heaters. AAAI 2026. [arXiv:2601.18123](#)**
 - Compared model-free RL against planning-based control for deadline-constrained domestic water heating under dynamic pricing, characterising where each approach wins and the cost of learning versus search.
- **Dixit, A., Khan, M.I., Ahmed, F., Brusey, J. Learning from Less: SINDy Surrogates in RL. ICLR 2025. [arXiv:2504.18113](#)**
 - Used sparse identification of nonlinear dynamics to build lightweight surrogate environment models, cutting the environment interactions required to train RL agents.

Under review / in preparation

- **Characterising and Fixing Numerical Instability in PPO with State-Dependent Exploration** (working title), targeting ICLR 2027.

RESEARCH EXPERIENCE

- **University College London (UCL)** Jun 2026 – Present
Research Software Developer, The Bartlett School of Sustainable Construction London, United Kingdom
 - Building the experimental software harness for a management research study on how large language models behave as agents within a simulated organisation, instantiating them in distinct workplace roles and logging behaviour for controlled comparison across conditions.
 - Designing multi-agent LLM pipelines with reproducible experiment configuration and structured logging, enabling researchers to run, replicate, and compare scenarios without engineering support.
 - Translating research questions from non-technical management academics into documented, maintainable Python, working directly with the principal investigator.
- **Coventry University (PhD research)** Sep 2024 – Present
Doctoral Researcher, CSMM Coventry, United Kingdom
 - Designing and training deep RL agents (PPO, SAC) for energy-efficient building heating control, covering reward design, observation-space selection, and training stability across a full experimental pipeline.
 - Benchmarking learned policies against planning-based control to establish where each approach is preferable under real-world pricing and comfort constraints.
 - Investigating the interpretability of learned control policies, extracting human-readable decision rules from trained agents and measuring fidelity against the policies they replace.
 - Diagnosed and fixed a reproducible numerical failure mode in a widely used PPO implementation, validated on continuous-control benchmarks.
 - Turing Scheme-funded international research visit to NUST, Pakistan (2026).
- **CureMD** Jul 2023 – Sep 2024
Associate AI Engineer Lahore, Pakistan
 - Production LLM/RAG systems: code assistant for 150+ developers (25%+ faster reviews), customer chatbot at 500+ daily queries (92% resolution accuracy), automated code grading with Llama 3.

TEACHING EXPERIENCE

- **Coventry University**

Teaching Assistant & Lab Demonstrator — *Deep Learning and Computer Vision (MSc)*

Aug 2025 – Jan 2026

Coventry, United Kingdom

- Weekly labs for 30+ MSc Data Science students; 200+ assignments graded per semester with structured formative feedback; collaborated with module leaders on curriculum refinement.

- **Coventry University**

Volunteer Teaching Assistant

2025 – 2026

Coventry, United Kingdom

- Supported module delivery and student learning beyond contracted duties.

Associate Fellow of the Higher Education Academy (AFHEA), Advance HE, 2026.

INVITED TALKS & PRESENTATIONS

- **Oral presentation**, AAAI 2026 — *Deadline-Aware, Energy-Efficient Control of Domestic Immersion Hot Water Heaters*.
- Talk, **Tri-University International AI Symposium** (Coventry University, NUST, Monash Suzhou), 2026.
- Progress Review Panel presentation, Coventry University, 2026.

ACADEMIC SERVICE

- Co-organiser, Tri-University International AI Symposium (Coventry–NUST–Monash Suzhou), Apr 2026.
- Peer reviewer for Elsevier and MDPI journals.

FUNDING, HONOURS & AWARDS

- **Turing Scheme Award** — funded international research visit 2026
- **Fully Funded PhD Studentship** — Coventry University, CSMM Sep 2024
- **2nd Runner-Up, Prime Minister's National Innovation Award** — Government of Pakistan, 40,000+ entries Jul 2023

TECHNICAL SKILLS

Languages: Python (expert), C++, MATLAB. **RL & Simulation:** Stable Baselines3, RLlib, Gymnasium, MuJoCo, BOPTEST, MCTS, SINDy. **ML/DL:** PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face. **Infrastructure:** Git, Docker, Linux, HPC/SLURM, Weights & Biases, LaTeX.

REFERENCES

Prof. James Brusey, Coventry University (PhD supervisor) — contact details on request.
Further references available upon request.